Materials Characterization Lab Notebook

Measurement & Analysis Record

This Notebook Belongs To

Lab / Organization

How to Use This Notebook

1. Record every measurement run immediately, in ink.
2. Capture the full setup: instrument, technique, parameters.
3. Log raw data file names, storage location and any deviations.
4. Have a witness sign off on critical or non-standard runs.

Materials Characterization Record

No. SAMPLE — DEMO

Date	Operator	Sample ID
2025-04-05	M. Wang	SLG-2025-018

Technique	Instrument
X-ray Diffraction (XRD)	Bruker D8 Discover

Sample Preparation

As-deposited thin film on Si(100), no further preparation.

Measurement Parameters

Parameter | Value | Note

Source: Cu Kα | λ=1.5406Å |
2θ range: 10-80° | Step: 0.02° | Scan speed: 2°/min

Environment (vacuum / ambient / temp)

Ambient, 22°C, 45% RH

Raw Data File & Location

Z:/Lab/XRD/2025-04/SLG-2025-018.raw

Observations

Strong (111) peak at 2θ=38.2°, FWHM=0.35°. Weak (200) peak detected.

Analysis / Results	Conclusion
Grain size (Scherrer): 24 nm. Lattice parameter: 4.08Å (PDF#04-0783 match).	Polycrystalline film, preferential (111) orientation.

Follow-up / Next Step

Cross-section TEM for grain morphology.

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